8.1 Introduction

In direct current and alternating current motors, starters are generally used for initial starting and safety purposes.

8.2 Direct current motor starters

8.2.1 Essential of starter

In general, all the motors, while starting, takes heavy starting current from the supply. Let us see the reason how the DC motors take heavy current during starting.

The armature current of the DC motor is given by the relation, I_a

$$I_a = \frac{V - E_b}{R_a}$$

V = line voltage

E_b = Back e.m.f or counter e.m.f

R_a = Armature resistance

When the motor is at rest, the induced back emf (E_b) in the armature is zero because the back emf is directly proportional to the speed of the motor, ($E_b \alpha N$). At starting time, the armature current is limited only by the resistance of the armature circuit. The armature resistance is very low and if full voltage is applied across the motor terminals, the armature will draw heavy current, since the armature current is inversely proportional to the resistance of armature circuit.

Example

10 H.P, 200 volt shunt motor has full load rated current 40A and armature resistance is about 0.2Ω. If the motor is directly switched on to supply, it would take armature current.

$$I_a = \frac{200 - 0}{0.2}$$

$$I_a = \frac{200}{0.2} = 1000\text{A}$$

At starting, the back emf is zero. The value of this current is 25 times the full load current. This high starting current may result in:

i. Large current flowing through the armature winding causing production of heat and it will damage the armature winding.
ii. Excessive sparking of the commutator.
iii. Excessive voltage drops in the line to which the motor is connected. Thus, the operation of the appliances connected to the same line, may refuse to work.

To avoid this excessive current at start, a variable resistance is connected in series with the armature circuit. This additional resistance limits the starting current. This creates the necessity of starters.

The starter has two basic functions.

i. Reducing starting current.
ii. Ensuring safe operation during next start.

8.2.2 Types of DC motor starter

i. Three point starter.
ii. Four point starter.

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